Y12 Physics — IB Diploma

Course Expectations • 2026–2027

Teacher	Mr. Alex Rawson • a.rawson@stleonards-fife.org
Room	PH1
We meet	Week A • Monday • Period 3 (11:15–12:10) Week A • Monday • Period 4 (12:15–13:05) Week A • Tuesday • Period 1 (08:55–09:50)
Course page	rawsonvault.com/physics/dp

1 What this course is

IB Diploma Physics, first of two years, on the guide first assessed in 2025, taken either as SL (150 hours, including 40 of practical work) or HL (240 hours, including 60). Exams are in May 2028.

The Diploma treats physics as five themes rather than a list of topics, and it asks something GCSE never did: that you can handle uncertainty honestly, argue from data, and say what your result actually shows. The step up from GCSE is real, and it lands in the first term. If it feels hard in October, that is the course working correctly — come and talk to me rather than concluding you are not a physicist.

2 What we cover

	Theme	Some of what is in it
A	Space, Time and Motion	Kinematics, forces and momentum, work energy and power; rigid body mechanics and relativity at HL
B	The Particulate Nature of Matter	Thermal energy transfers, the greenhouse effect, gas laws, current and circuits; thermodynamics at HL
C	Wave Behaviour	Simple harmonic motion, the wave model, wave phenomena, standing waves and resonance, the Doppler effect
D	Fields	Gravitational, electric and magnetic fields, and induction
E	Nuclear and Quantum Physics	Structure of the atom, radioactive decay, fission and fusion, and the quantum world

Running through all five are the **tools** — experimental technique, uncertainty and data analysis, the mathematics — and the **nature of science**. They are not a separate unit; they are assessed inside everything else.

3 What you need

Every lesson

- A4 notebook or folder — DP physics generates a lot of worked problems
- Scientific calculator — on the IB permitted list
- The IB Physics data booklet — learn your way around it now, not in May 2028
- Pencil, pen, ruler

Keep it organised in a ring binder with dividers

When we need them

- Lab notebook for the practical programme
- A laptop for data analysis and the internal assessment

4 How you are assessed

Weight	Component	What it is
36%	Paper 1	1A multiple choice, and 1B data analysis
44%	Paper 2	Structured and extended response
20%	Internal assessment	The scientific investigation — your own

The internal assessment is one independent investigation you design, run, and write up in around 3,000 words, drawn from roughly ten hours of work. It is 20% of your grade and it is the only part you control completely. We start it early in the course; the internal deadlines are not moveable, because the school's submission date to the IB is not moveable.

The **practical programme** — 40 hours at SL, 60 at HL — is a course requirement in its own right, not just IA preparation.

5 Homework

Expect two pieces a week, 45–60 minutes each: problem sets, past-paper questions, and reading ahead so lessons can be spent on the hard parts rather than the definitions.

Reading ahead is not optional in this course. Come having met the vocabulary, and the lesson becomes about the physics.

6 Practical work and the investigation

You will keep a lab notebook for the whole practical programme. Record uncertainties as you take the reading, not afterwards — an uncertainty invented at the writing-up stage is fiction, and the IB is very good at spotting fiction.

Safety. Standard lab rules, and the same one that matters most: tell me immediately if something breaks or goes wrong. You are not in trouble for reporting it.

On the investigation: choose something you can actually measure with the equipment we have, and choose it early. Almost every weak IA is a good idea that ran out of time.

7 Deadlines and late work

Deadlines are given in class and posted on the course page. Work handed in late is marked and returned, but recorded as late; more than a week late is recorded as not submitted.

If something is going wrong — illness, deadlines stacking up, anything at home — **come and tell me before the deadline.** An extension asked for in advance is nearly always fine. Asking afterwards is different — by then, I usually cannot say yes.

8 Absence

Being away does not remove the work. It only delays it. **It is your job to find out what you missed** — check the course page first, then ask me.

- **Lessons.** Notes and resources are on the course page. Catch up before the next lesson.
- **Assessments.** Take the assessment within three school days of coming back. Come and book a time yourself — this is your responsibility, and I will not remind you.
- **Extended absence.** See me on your first day back and we will agree a realistic plan and a set of dates.

9 Phones, laptops, and AI

Phones go straight in your locker when you arrive. They stay there for the whole school day, in every lesson.

Laptops are open when the task needs them and closed when it does not. I will say which.

AI tools are genuinely useful and you should learn to use them well. In this course:

- **Fine:** asking it to explain something you are stuck on, generating extra practice questions, checking your own reasoning after you have done the work yourself.
- **Not fine:** getting it to produce an answer, a paragraph, a method or code that you then hand in as your thinking.
- **Always:** write one line at the end saying what you used and what for. Declaring it is never penalised. Not declaring it is a problem.

The reason is simple. The assessment has no AI in it. The moment you actually need to understand something has no AI in it either. If an AI does the work for you, you lose the time you needed to get better at physics.

10 Academic honesty

You may work together — you learn faster when you do. But **the writing must be yours, done on your own.**

- Write down who you worked with, on the work itself.
- Cite anything you used — book, website, video, diagram, dataset. Putting it in your own words still needs a citation: you are borrowing the idea even when you change the words.
- The IB treats malpractice on internal assessment as a Diploma-level matter, not a class-level one. Your IA is submitted with a declaration that the work is your own, and that declaration is not something I can soften on your behalf.

11 In the classroom

- Arrive on time and ready to start.
- Ask the question. Someone else has it too, and I would rather stop the lesson than let you fall behind.
- Wrong answers said out loud are how the lesson works. Nobody is laughed at in here.
- Leave the lab ready for the next class.

12 How to do well in this course

Do the problems. All of them, before the lesson where we go through them — watching a solution is not the same as producing one, and the exam asks you to produce.

Get comfortable with uncertainty early; it is the single thing that separates a strong DP physicist from a strong GCSE one. Start the IA on time. And ask in October, not in February: this course builds, and the people who struggle in Year 13 are almost always the ones who let a first-term gap go quiet.